

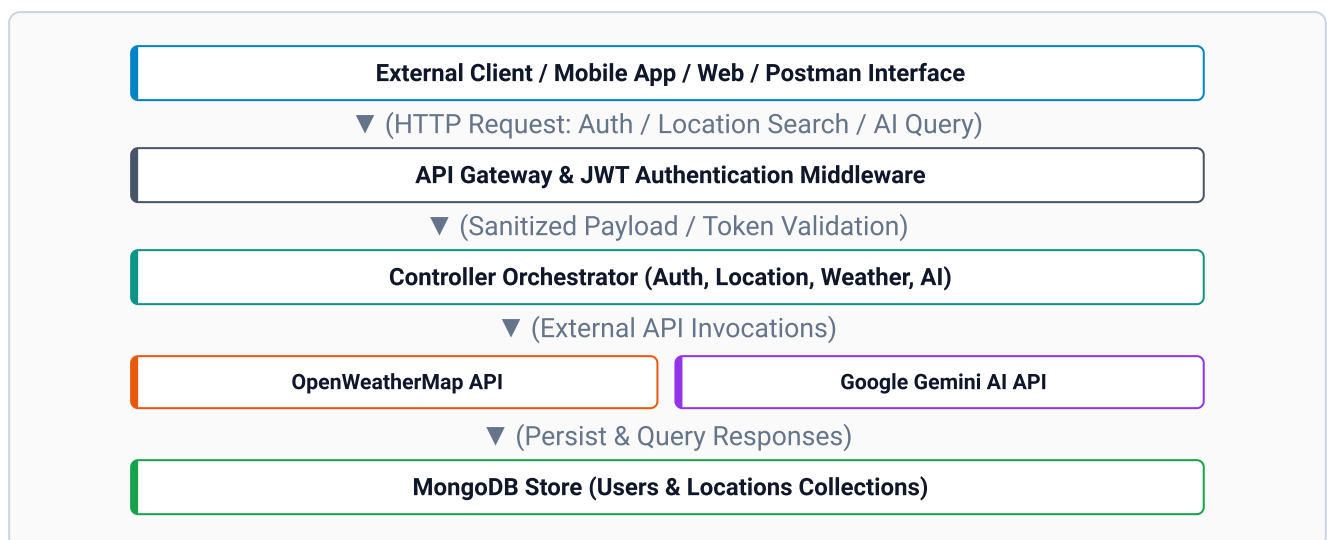
PROJECT DESIGN PHASE-II

Data Flow Diagrams & User Stories

Date:	29/09/26	Team ID:	SWTID-2026-8956
Project Name:	WeatherWise AI	Maximum Marks:	4 Marks
Team Size:	4 Members		
Team Members:	SANMATHI KUMAR (Leader), ARIFA ZOHRA, AINUL LIYANA P M, VAISHALI S		

1. DATA FLOW DIAGRAMS (DFD) OVERVIEW

A Data Flow Diagram (DFD) visually maps information flow across the WeatherWise AI system. It details data entry points, transformation routes through controller workflows, third-party API processing, and database storage updates.



2. EXECUTION SEQUENCE

- Step 1:** User issues registration/login requests or weather queries via secure API routes.
- Step 2:** JWT Middleware validates bearer signatures and passes sanitized input to targeted Controllers.
- Step 3:** Weather Controller fetches live metrics from OpenWeatherMap API.
- Step 4:** Metrics are sent to Google Gemini AI API to construct natural language clothing/activity suggestions.
- Step 5:** Standardized JSON payloads are delivered to the user and saved in MongoDB.

3. USER STORIES & ACCEPTANCE CRITERIA

Below is the complete set of User Stories, functional Epics, acceptance criteria, and release sprint schedules for WeatherWise AI:

User Type	Epic	ID	User Story / Task	Acceptance Criteria	Priority	Release
Registered User	Registration	USN-1	As a user, I can register using my name, email, and password.	Password min 6 chars; unique email validated & saved lowercase.	High	Sprint-1
Registered User	Authentication	USN-2	As a user, I can log in to obtain an authentication token.	Returns valid JWT session token on success.	High	Sprint-1
Registered User	Locations	USN-3	As a user, I can save my favorite cities to my profile.	Compound index prevents duplicate city records per user.	Medium	Sprint-1
Registered User	AI Insights	USN-4	As a user, I can request tailored weather & attire recommendations.	Returns AI natural language summary based on real-time metrics.	High	Sprint-2
Public User	Weather View	USN-5	As a guest, I can query basic weather data without logging in.	GET `/api/weather/:city` returns raw metrics publicly.	Low	Sprint-2
Administrator	System Admin	USN-6	As an admin, I can monitor DB integrity and service uptime.	Access health endpoints and log metrics.	High	Sprint-2